

The five shortcut bots

Seeds 42-46, with Pattern Detection updated to sequential overlapping 4-3-2.

5 shortcut bots Ten unique pairings	50,000 tournament games Five seeds combined	2nd Pattern Detection rank 10,097 game wins	51.0% 4-move round win rate When that width was selected
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This report includes only Regular, Aggressive, Short-Term Memory, Random, and the updated Pattern Detection. Pattern Detection means the sequential 4-move, then 3-move, then 2-move search, counting every overlapping match at the first successful width. It uses Regular during warmup and when all widths miss. The memory window remains fifteen actual opponent moves.

Aggressive leads by total game wins: 10,303 wins, 6,786 losses, and 2,911 ties. Pattern Detection is second with 10,097 wins, 6,397 losses, and 3,506 ties. Each bot plays 20,000 games and 100,000 rounds against the other four bots across all five seeds.

Pattern Detection wins its combined matchup against each of the other four bots, including Aggressive, 2,136-1,849-1,015. It also has the largest game margin (+3,700) and round margin (+12,649). Aggressive's higher win total reflects stronger results elsewhere in this small field, especially against Short-Term Memory; winning every opponent matchup does not guarantee first place by total wins.

The outcome analysis covers Pattern Detection's 100,000 rounds. Four-move choices have the highest observed round win rate, 51.0%, followed by three moves (37.4%) and two (36.9%). Excluding Short-Term Memory drops the four-move rate to 37.5%. The longer search is particularly successful against the most exploitable opponent, so its headline rate should be read with that exposure in mind.

Same rules, narrower field. The ten pairings were taken from the saved five-seed runs and replayed for exact verification. Every standing and success rate here uses only these five bots. No five-move search, pooled pattern rule, or ensemble adviser is part of this report. Regular remains the shortcut main mode.

What each shortcut bot does

The report name Pattern Detection always refers to sequential overlapping 4-3-2 + Regular fallback.

Bot	Learning account	How it chooses
Regular	Opponent moves on ties and losses, separately for each round slot.	Counter the most frequent recorded move. Randomly choose among tied maxima; uniform random if the account is empty.
Aggressive	Regular counts, plus one extra count on a pure loss. Losses count twice and ties once.	Counter the most frequent recorded move, resolving tied maxima randomly.
Short-Term Memory	Newest ten opponent moves from ties and losses, separately for each round slot. Wins do not enter the list.	Counter the most frequent move in that slot's retained list, resolving tied maxima randomly.
Random	Regular frequency counts plus Flux: each move starts with one Flux count and gains Flux on wins.	Draw a move with weights from the combined frequency and Flux counts. This is weighted random, not necessarily a uniform one-third draw.
Pattern Detection	Latest fifteen actual opponent moves across all round slots; Regular also maintains its per-slot tie/loss account.	Search 4, then 3, then 2 only after larger widths miss. Count overlaps and duplicate counter tickets. Draw one ticket from the first successful width; use Regular on warmup or no match.

Pattern search in a newest-first history

A 4-move match uses the latest three moves as context. The three older entries of a historical four-move window must equal that context; its newest entry is the historical next move, whose counter becomes a ticket. Three moves uses a two-move context; two uses one. Every start advances by one, including after a match.

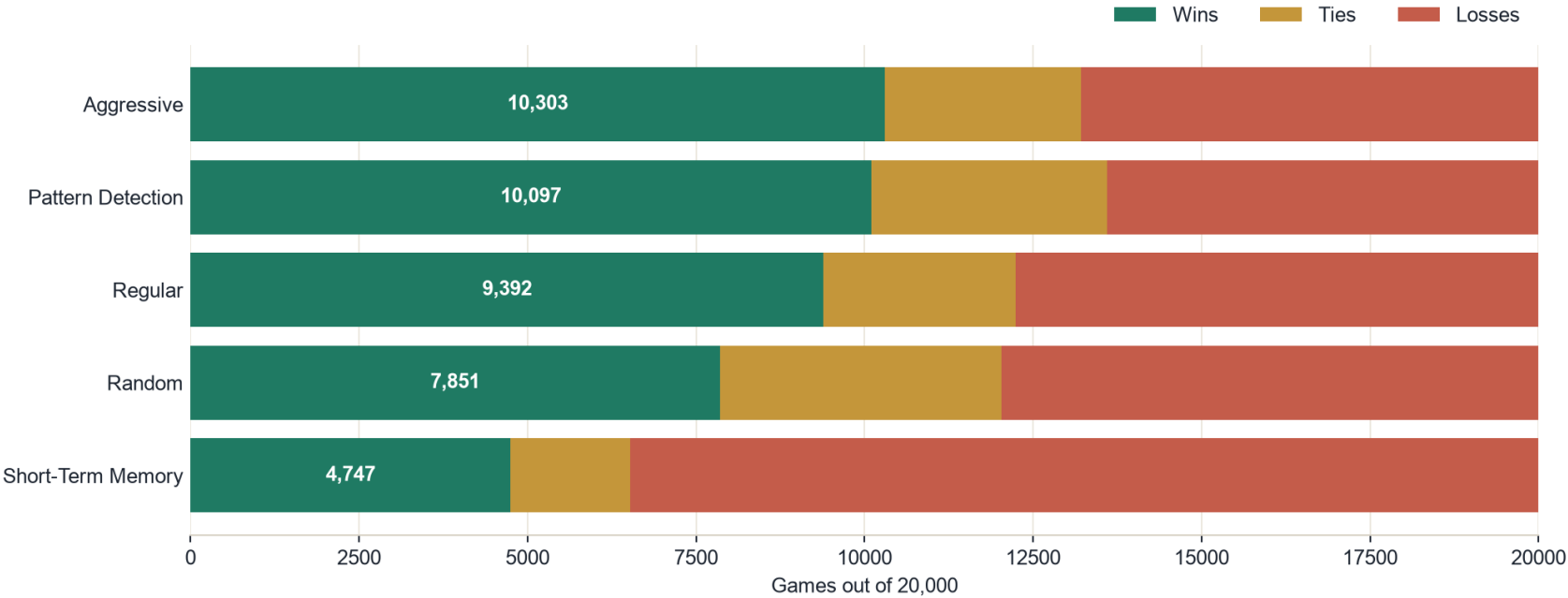
Search	Context length	Maximum windows in fifteen moves
4-move	3 moves	12
3-move	2 moves	13
2-move	1 move	14

All comparisons start with empty accounts for each opponent and seed. Memory persists across the 1,000 games within that pairing. This report evaluates the simulator's updated rule; it does not certify or edit the shared Apple Shortcut implementation.

Combined standings

Rank is game wins, then fewer losses. Tied games remain in Win %. Decisive % excludes ties. All counts combine five seeds.

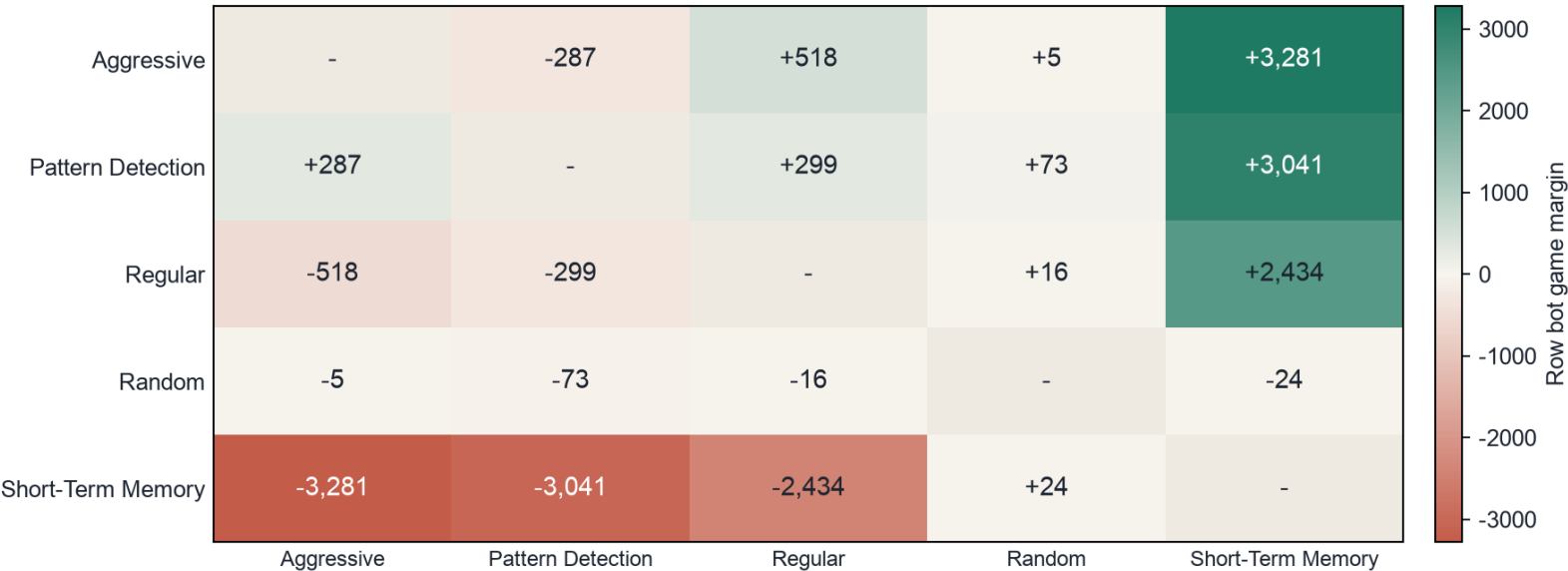
#	Bot	Game W	Game L	Game T	Win %	Margin	Decisive %	Round W	Round L	Round T
1	Aggressive	10,303	6,786	2,911	51.5%	+3,517	60.3%	46,172	35,753	18,075
2	Pattern Detection	10,097	6,397	3,506	50.5%	+3,700	61.2%	44,047	31,398	24,555
3	Regular	9,392	7,759	2,849	47.0%	+1,633	54.8%	43,078	38,132	18,790
4	Random	7,851	7,969	4,180	39.3%	-118	49.6%	33,439	33,834	32,727
5	Short-Term Memory	4,747	13,479	1,774	23.7%	-8,732	26.0%	30,371	57,990	11,639



All ten combined matchups

Each row combines five 1,000-game runs. Counts are from Bot A's side: 5,000 games and 25,000 rounds per row.

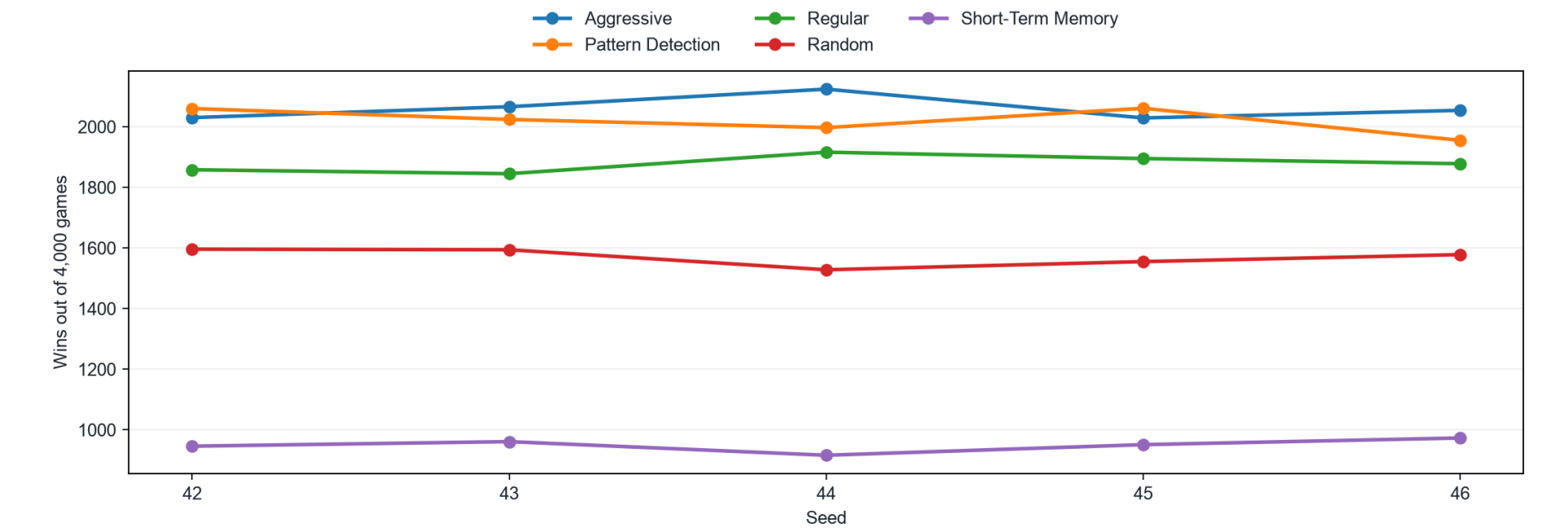
Bot A	Bot B	Game W	Game L	Game T	Margin	Round W	Round L	Round T
Aggressive	Pattern Detection	1,849	2,136	1,015	-287	8,869	9,590	6,541
Aggressive	Regular	2,423	1,905	672	+518	11,821	10,417	2,762
Aggressive	Random	1,971	1,966	1,063	+5	8,433	8,452	8,115
Aggressive	Short-Term Memory	4,060	779	161	+3,281	17,049	7,294	657
Pattern Detection	Regular	2,160	1,861	979	+299	9,177	8,463	7,360
Pattern Detection	Random	2,007	1,934	1,059	+73	8,398	8,210	8,392
Pattern Detection	Short-Term Memory	3,794	753	453	+3,041	16,882	5,856	2,262
Regular	Random	1,984	1,968	1,048	+16	8,531	8,385	8,084
Regular	Short-Term Memory	3,642	1,208	150	+2,434	15,667	8,749	584
Random	Short-Term Memory	1,983	2,007	1,010	-24	8,392	8,472	8,136



How the five seeds compare

Every seed column shows game wins and rank in parentheses. Each bot plays 4,000 games per seed; combined ranking uses summed wins.

Bot	Seed 42	Seed 43	Seed 44	Seed 45	Seed 46	Combined W
Aggressive	2,030 (2)	2,066 (1)	2,124 (1)	2,029 (2)	2,054 (1)	10,303
Pattern Detection	2,060 (1)	2,024 (2)	1,997 (2)	2,061 (1)	1,955 (2)	10,097
Regular	1,858 (3)	1,845 (3)	1,916 (3)	1,895 (3)	1,878 (3)	9,392
Random	1,596 (4)	1,594 (4)	1,528 (4)	1,555 (4)	1,578 (4)	7,851
Short-Term Memory	946 (5)	961 (5)	916 (5)	951 (5)	973 (5)	4,747

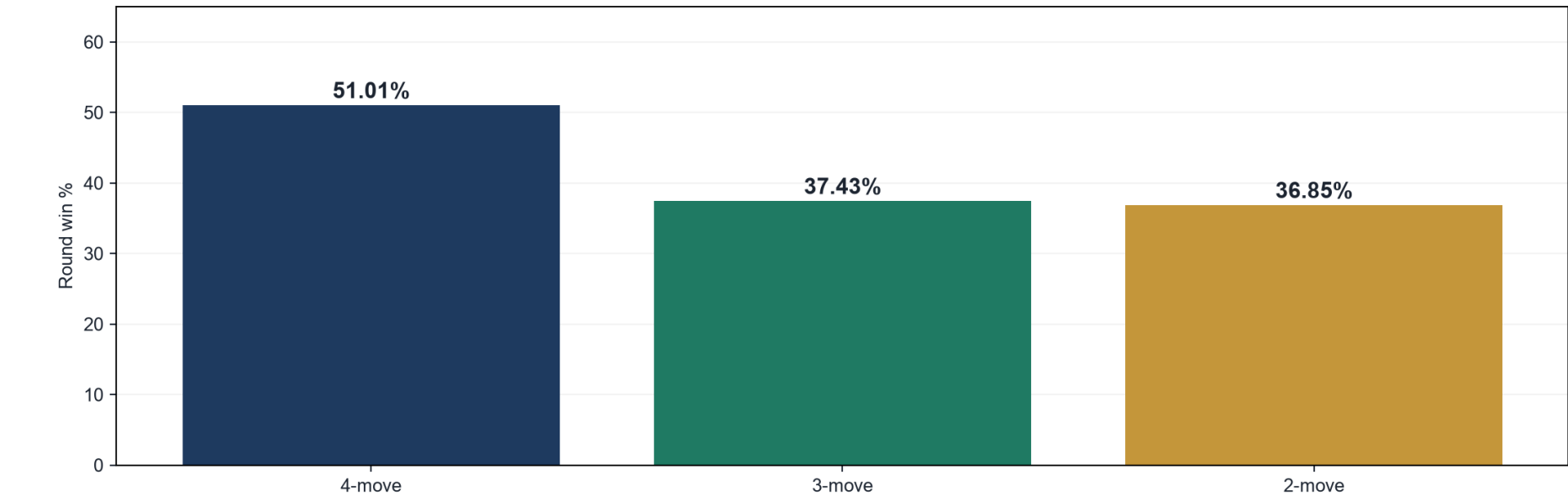


Small changes in rank between seeds do not change the rules. The complete 50 pairing records are retained in the saved analysis JSON.

How successful were the selected patterns?

Actual round outcomes for Pattern Detection only, across its four opponents and five seeds. Width counts are exclusive: the first successful width supplied the played ticket.

Choice source	Rounds	Wins	Losses	Ties	Round win %	Decisive %	Net score %
4-move	49,961	25,487	14,680	9,794	51.0%	63.5%	+21.63%
3-move	31,527	11,799	10,474	9,254	37.4%	53.0%	+4.20%
2-move	16,643	6,133	5,623	4,887	36.9%	52.2%	+3.06%
Regular fallback	1,569	544	527	498	34.7%	50.8%	+1.08%
Regular warmup	300	84	94	122	28.0%	47.2%	-3.33%
ALL	100,000	44,047	31,398	24,555	44.0%	58.4%	+12.65%



Four moves leads the raw win rate and net score. Three moves is tried only after four misses, and two only after both miss; these buckets therefore describe different historical situations. Round win rate includes ties. Net score is (wins - losses) / all rounds. These are not game win rates.

Opponent exposure and pattern success

The table shows round win % / selected round count. Four-move success is especially high against Short-Term Memory.

Opponent	4-move: % / n	3-move: % / n	2-move: % / n	Fallback: % / n
Aggressive	39.94% / 10,990	38.14% / 8,592	35.74% / 4,882	33.84% / 461
Regular	37.78% / 10,320	36.27% / 9,328	35.72% / 5,011	31.58% / 266
Random	34.24% / 9,120	33.11% / 10,635	33.41% / 5,067	40.78% / 103
Short-Term Memory	72.07% / 19,531	54.44% / 2,972	53.77% / 1,683	35.45% / 739

Exclude Short-Term Memory as a sensitivity check

The following subgroup keeps only Regular, Aggressive, and Random. Removing the weakest opponent helps show how much the raw rates depend on an exploitable opponent, but does not hold the selected histories constant.

Width	All four opponents win %	Other three win %	Other three loss %	Other three net %	Other three rounds
4-move	51.01%	37.50%	34.3%	+3.17%	30,430
3-move	37.43%	35.65%	33.5%	+2.11%	28,555
2-move	36.85%	34.95%	34.5%	+0.41%	14,960

Four moves is a more specific context, while shorter patterns have more opportunities to match. A higher observed rate can reflect the opponent, repetitive behavior, and when the fallback ladder reaches that width. The same-history comparison on the next page provides a more comparable set of opportunities.

Compare widths on the same matching histories

Every four-move match implies shorter matches too. All three widths can therefore be evaluated on exactly the same rounds where four matched.

Width	All four expected win %	All four expected net %	Other three expected win %	Other three expected net %
4-move	51.03%	+21.67%	37.56%	+3.30%
3-move	50.03%	+19.51%	36.52%	+1.12%
2-move	49.14%	+17.91%	35.61%	-0.26%

All-four comparison: 49,961 recorded rounds. Other-three comparison: 30,430 rounds, excluding Short-Term Memory. Within each subgroup, every width uses the same history, committed opponent move, and denominator.

Exact ticket probabilities, with no extra played moves

Each matching historical window contributes one counter ticket. For a recorded round, score every ticket at each width against the actual opponent move, then divide by that width's ticket count. Summing these probabilities gives the expected round win, loss, and tie counts for a uniform ticket draw. Duplicates and overlapping samples remain. The calculation consumes no random numbers and changes no learning.

One-round comparison, not a replacement tournament

These hypothetical outcomes describe the recorded sequential 4-3-2 trajectories. Actually choosing another width could change subsequent learning and opponent behavior. The comparison supports an interpretation of context on common opportunities; it does not establish whole-game performance for a bot forced to use one width.

How many matching samples supported the choice?

Selected width	One sample: rounds / win %	Two samples: rounds / win %	Three or more: rounds / win %
4-move	27,077 / 40.31%	8,490 / 43.43%	14,394 / 75.62%
3-move	16,716 / 37.47%	9,998 / 37.26%	4,813 / 37.63%
2-move	1,824 / 37.55%	2,754 / 37.62%	12,065 / 36.57%

More samples are not automatically more independent evidence: overlapping windows can share moves, and easy opponents can create repeated matches.

Pattern outcomes by seed and verification

Seed	4-move win %	3-move win %	2-move win %	Fallback win %	Pattern game W-L-T
42	51.71%	38.05%	35.26%	37.77%	2,060-1,258-682
43	51.18%	37.35%	37.56%	33.53%	2,024-1,278-698
44	51.65%	37.27%	36.06%	35.84%	1,997-1,305-698
45	51.60%	36.91%	37.28%	32.36%	2,061-1,260-679
46	48.89%	37.55%	38.09%	33.88%	1,955-1,296-749

The ten pairings and five seeds

The saved seed runs were restricted to these five bots before any standing or rate was calculated. Every seed retains the original A/B order and starts a fresh seeded RNG for each opponent pairing. The ten matches at every seed were replayed with the unchanged simulator. All six game and round counts matched the saved results exactly: 50 verified pairing runs in total.

Observing the updated Pattern Detection

An observer records both committed moves before either bot learns. An independent overlapping scanner identifies whether 4, 3, or 2 supplied the first matching ticket set, or whether the move used Regular warmup/fallback. It consumes no RNG and changes no account. Twenty observed Pattern Detection pairings reproduce their saved outcomes. Every width count agrees with the simulator's usage counters; the buckets sum to 44,047 wins, 31,398 losses, and 24,555 ties across 100,000 rounds.

Saved sources and reproducibility

shortcut_five_5seeds_42-46.json stores all 50 pairing records, combined standings, pattern outcomes by seed and opponent, matching sample groups, and same-history expected outcomes. analyze_shortcut_five.py reproduces the restricted analysis using results_23bot_5seeds_42-46.json and rps_bots.py. Older result files, the source script, and earlier PDFs retain their hashes. The shortcut was not run or changed. Report date: 1 October 2026.

This report describes simulated opponents and five seeds. Games in a pairing share memory, and each seed reuses its starting RNG state across pairings. Small rate differences should not be treated as independent-game statistical proof. The field and updated pattern rule are held fixed throughout this report.

Every bot against its four opponents

Combined records from each bot's own side. Each opponent pairing is 5,000 games and 25,000 rounds.

1. Aggressive

Opponent	Game W	Game L	Game T	Game win %	Round W	Round L	Round T
Pattern Detection	1,849	2,136	1,015	37.0%	8,869	9,590	6,541
Regular	2,423	1,905	672	48.5%	11,821	10,417	2,762
Random	1,971	1,966	1,063	39.4%	8,433	8,452	8,115
Short-Term Memory	4,060	779	161	81.2%	17,049	7,294	657

2. Pattern Detection

Opponent	Game W	Game L	Game T	Game win %	Round W	Round L	Round T
Aggressive	2,136	1,849	1,015	42.7%	9,590	8,869	6,541
Regular	2,160	1,861	979	43.2%	9,177	8,463	7,360
Random	2,007	1,934	1,059	40.1%	8,398	8,210	8,392
Short-Term Memory	3,794	753	453	75.9%	16,882	5,856	2,262

3. Regular

Opponent	Game W	Game L	Game T	Game win %	Round W	Round L	Round T
Aggressive	1,905	2,423	672	38.1%	10,417	11,821	2,762
Pattern Detection	1,861	2,160	979	37.2%	8,463	9,177	7,360
Random	1,984	1,968	1,048	39.7%	8,531	8,385	8,084
Short-Term Memory	3,642	1,208	150	72.8%	15,667	8,749	584

Every bot against its four opponents (continued)

Combined records from each bot's own side. Each opponent pairing is 5,000 games and 25,000 rounds.

4. Random

Opponent	Game W	Game L	Game T	Game win %	Round W	Round L	Round T
Aggressive	1,966	1,971	1,063	39.3%	8,452	8,433	8,115
Pattern Detection	1,934	2,007	1,059	38.7%	8,210	8,398	8,392
Regular	1,968	1,984	1,048	39.4%	8,385	8,531	8,084
Short-Term Memory	1,983	2,007	1,010	39.7%	8,392	8,472	8,136

5. Short-Term Memory

Opponent	Game W	Game L	Game T	Game win %	Round W	Round L	Round T
Aggressive	779	4,060	161	15.6%	7,294	17,049	657
Pattern Detection	753	3,794	453	15.1%	5,856	16,882	2,262
Regular	1,208	3,642	150	24.2%	8,749	15,667	584
Random	2,007	1,983	1,010	40.1%	8,472	8,392	8,136